

**LLM-POWERED HYBRID MOVIE RECOMMENDATION SYSTEM USING
GENERATIVE AI**

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**LLM-POWERED HYBRID MOVIE RECOMMENDATION SYSTEM
USING GENERATIVE AI**

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LLM-POWERED HYBRID MOVIE RECOMMENDATION SYSTEM USING GENERATIVE AI

ABSTRACT

This research addresses the persistent limitations of existing recommender systems by proposing a hybrid movie recommendation framework that integrates predictive modeling, traditional filtering, and Large Language Model (LLM)-powered query interpretation. While most prior approaches have emphasized accuracy, they often neglect other critical aspects such as novelty, diversity, and explainability, which are essential for enhancing user trust and engagement. To overcome these gaps, the study combines multiple techniques within a unified architecture. Predictive modeling using Random Forest and XGBoost is employed to forecast movie popularity and ratings. Content-based similarity (cosine similarity, embeddings) and popularity-based filtering (weighted ratings and vote counts) are incorporated through a weighted fusion strategy. In parallel, LLMs (e.g., GPT-4, LLaMA-2, Gemini) supported by tools such as Hugging Face are used to interpret free-form user queries, extract entities, and detect sentiment, enabling context-aware and narrative-driven recommendations. Generative AI further enriches incomplete metadata and generates natural language explanations, improving transparency and user experience. The framework will be evaluated using metrics such as Precision@K, Recall@K, novelty, diversity, and long-tail bias indicators. Comparative analysis will benchmark performance against collaborative filtering, content-based filtering, and LLM-only baselines. A functional prototype will be deployed using a FastAPI or Flask backend integrated with LLMs and served via a Streamlit frontend, demonstrating the practical feasibility of the approach. This study is expected to contribute to the design of next-generation recommender systems that are accurate, personalized, fair, and transparent. Beyond movies, the framework has potential applications in e-commerce, education, and digital media platforms, where user engagement, fairness, and explainability are equally critical.

Keywords: Hybrid Recommender Systems, Predictive Modeling, Large Language Models, Generative AI, Personalization.

SISTEM CADANGAN FILEM HIBRID BERKUASA LLM MENGGUNAKAN AI GENERATIF

ABSTRAK

Penyelidikan ini menangani had berterusan sistem pengesyor sedia ada dengan mencadangkan rangka kerja pengesyoran filem hibrid yang menyepadukan pemodelan ramalan, penapisan tradisional dan tafsiran pertanyaan dikuasakan Model Bahasa Besar (LLM). Walaupun kebanyakan pendekatan terdahulu telah menekankan ketepatan, mereka sering mengabaikan aspek kritikal lain seperti kebaruan, kepelbagaian dan kebolehjelasan, yang penting untuk meningkatkan kepercayaan dan penglibatan pengguna. Untuk mengatasi jurang ini, kajian ini menggabungkan pelbagai teknik dalam seni bina bersatu. Pemodelan ramalan menggunakan Random Forest dan XGBoost digunakan untuk meramalkan populariti dan rating filem. Persamaan berasaskan kandungan (persamaan kosinus, benam) dan penapisan berasaskan populariti (penilaian wajar dan kiraan undi) digabungkan melalui strategi gabungan berwajaran. Secara selari, LLM (cth., GPT-4, LLaMA-2) yang disokong oleh alatan seperti spaCy, Hugging Face digunakan untuk mentafsir pertanyaan pengguna bentuk bebas, mengekstrak entiti dan mengesan sentimen, mendayakan pengesyoran berasaskan konteks dan didorong naratif. AI Generatif memperkayakan lagi metadata yang tidak lengkap dan menjana penjelasan bahasa semula jadi, meningkatkan ketelusan dan pengalaman pengguna. Rangka kerja akan dinilai menggunakan metrik seperti Precision@K, Recall@K, kebaruan, kepelbagaian dan penunjuk berat sebelah ekor panjang. Analisis perbandingan akan menanda aras prestasi terhadap penapisan kolaboratif, penapisan berasaskan kandungan dan garis dasar LLM sahaja. Prototaip berfungsi akan digunakan menggunakan bahagian belakang FastAPI atau Flask yang disepadukan dengan LLM dan disampaikan melalui bahagian hadapan Streamlit, menunjukkan kebolehlaksanaan praktikal pendekatan tersebut. Kajian ini dijangka menyumbang kepada reka bentuk sistem pengesyor generasi seterusnya yang tepat, diperibadikan, adil dan telus. Selain filem, rangka kerja ini mempunyai aplikasi yang berpotensi dalam e-dagang, pendidikan dan platform media digital, di mana penglibatan pengguna, keadilan dan kebolehjelasan adalah sama kritikal.

Kata kunci: Sistem Pengesyor Hibrid, Pemodelan Ramalan, Model Bahasa Besar, AI Generatif, Pemberibadian.

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LIST OF SYMBOLS AND ABBREVIATIONS

LLM	:	Large Language Model
GAN	:	Generative Adversarial Network
VAE	:	Variational Autoencoder
DBN	:	Deep Belief Network
MBO	:	Monarch Butterfly Optimization
CF	:	Collaborative Filtering
API	:	Application Programming Interface
TF-IDF	:	Term Frequency–Inverse Document Frequency
OSEMN	:	Obtain, Scrub, Explore, Model, and Interpret
RMSE	:	Root Mean Square Error
MAE	:	Mean Absolute Error
F1	:	F1 Score (harmonic mean of precision and recall)
IMDB	:	Internet Movie Database
OMDB	:	Open Movie Database

CHAPTER 1: INTRODUCTION

1. Research Background

Recommender systems have become central to modern digital platforms, driving personalized access to movies, e-commerce products, and diverse forms of digital media. Traditional approaches, including collaborative filtering and hybrid methods, have demonstrated strong performance in modeling user–item interactions and generating accurate recommendations (Mahmood et al., 2022). Despite these successes, such methods remain constrained by their reliance on structured inputs and are often challenged by data sparsity and the cold-start problem, which limit their effectiveness for new users or items. Recent advances in deep learning have contributed significantly to overcoming some of these challenges by enabling richer feature representation and more robust predictive modeling. Techniques such as optimized deep belief networks and transformer-based architectures have reported substantial improvements in predictive accuracy (Sridhar et al., 2023; Siet et al., 2024). In parallel, generative AI approaches, including Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and autoencoder-based models have been introduced to augment metadata, alleviate sparsity, and generate synthetic representations, thereby further enhancing recommendation quality (Ayemowa et al., 2024; Ji et al., 2023).

Alongside these developments, Large Language Models (LLMs) have opened a new dimension in recommender system design by enabling conversational and narrative-driven recommendations. Their capacity to interpret natural language queries allows systems to capture contextual preferences, emotional tone, and narrative cues that extend beyond traditional metadata (Sun et al., 2024; Eberhard et al., 2024). However, despite these advances, current LLM-based recommenders still face persistent limitations, particularly in achieving consistent personalization, ensuring novelty and diversity, and mitigating fairness concerns (Lichtenberg et al., 2023). These enduring gaps underscore the importance of developing hybrid frameworks that combine the predictive strength of machine learning models, the contextual richness of LLM-based query interpretation, and the adaptability of generative AI. Such integration provides a pathway toward recommender systems that not only achieve high levels of accuracy but also advance novelty, diversity, and explainability, thereby addressing both technical and user-centric challenges.

1.1. Problem Statement

Both predictive modeling and LLM-powered recommenders demonstrate strong potential, their integration into a cohesive recommendation pipeline remains limited. Predictive models improve personalization by forecasting popularity and ratings, but they depend heavily on metadata quality and rarely extend to conversational contexts (Sridhar et al., 2023; Siet et al., 2024). In contrast, LLM-based systems, while effective at interpreting free-form queries and generating explanations, often underperform in personalization, novelty, and bias mitigation, and face challenges of scalability and latency (Sun et al., 2024; Ji et al., 2023). Most existing hybrid or LLM-driven frameworks prioritize accuracy but fall short on explainability, novelty, and diversity, the factors that are critical for maintaining user trust and engagement in dynamic streaming environments (Eberhard et al., 2024; Lichtenberg et al., 2023). In addition, traditional hybrid models continue to struggle with cold-start issues, static metadata reliance, and lack of transparent reasoning (Mahmood et al., 2022; Ayemowa et al., 2024).

This research therefore addresses these weaknesses by proposing a unified framework that combines predictive modeling, LLM-based query interpretation, and hybrid filtering. The system aims to deliver recommendations that are not only accurate but also diverse, novel, explainable, and context-aware, bridging the gap between structured predictive methods and conversational, user-centric experiences (Ayemowa et al., 2024; Sun et al., 2024).

1.2. Research Questions

1. Which features strongly influence movie popularity and user ratings in predictive modeling?
2. How can LLM-based query interpretation be integrated with predictive modeling to enhance recommendation effectiveness?
3. How does the hybrid framework perform compared to traditional and LLM-only methods across standard recommendation metrics?

1.3. Research Objectives

1. To identify key features influencing movie popularity and user ratings for predictive modeling.
2. To develop a hybrid recommendation framework that integrates LLM-based query interpretation with predictive modeling.

3. To evaluate the framework through comparative analysis using standard recommendation metrics.

1.4 Scope and Contribution

1.4.1 Scope

This research focuses on the design and evaluation of a hybrid movie recommendation framework that integrates predictive modeling with Large Language Model (LLM)-based query interpretation. The study makes use of structured movie metadata (genres, ratings, cast, crew, and summaries) along with free-form natural language queries to address the following key areas:

1. **Predictive Modeling:** Application of classification and regression models (e.g. Random Forest, Gradient Boosting) to forecast movie popularity and ratings, providing a baseline for personalization.
2. **Natural Language Query Interpretation:** Use of advanced LLMs (e.g., GPT-4, LLaMA, Gemini) to extract structured preference profiles (genres, moods, time frames) from user queries, enabling context-aware recommendations.
3. **Hybrid Recommendation Framework:** Integration of predictive outputs with content-based similarity and popularity-based filtering, combined through weighted fusion to optimize accuracy, diversity, and novelty.
4. **Bias Mitigation and Explainability:** Application of niche-boosting and generative methods to reduce popularity bias, enrich incomplete metadata, and generate natural language explanations for recommendations.
5. **Evaluation and Deployment:** Comparative assessment against traditional baselines using metrics such as Precision@K, Recall@K, novelty, and diversity, followed by prototype deployment with FastAPI/Flask backend and a Streamlit interface.

1.4.2 Contribution

This study contributes to recommender system research by presenting a unified hybrid framework that bridges predictive modeling, traditional filtering methods, and LLM-based query interpretation. It advances personalization through predictive baselines, enhances user experience through conversational and context-aware recommendations, and strengthens transparency via explainable outputs and enriched metadata. By incorporating bias mitigation strategies, the system improves diversity and novelty in suggestions, addressing common shortcomings of existing models. Furthermore, the deployment of a working prototype demonstrates the practical applicability of the framework in real-world streaming platforms. Collectively, these contributions establish a data-driven, explainable, and user-centric recommendation approach that extends beyond academic experimentation to practical deployment.

CHAPTER 2: LITERATURE REVIEW

2.1 Traditional and Hybrid Approaches

Early work in recommender systems primarily leveraged structured metadata and user–item interaction patterns. Hybrid approaches that combine collaborative filtering with other techniques have demonstrated improved coverage and accuracy by capturing complementary signals. For example, Mahmood et al. (2022) developed a hybrid movie recommender that integrated collaborative filtering with association rule mining; collaborative filtering captured user–item affinities, while association rules surfaced latent item associations, together improving recommendation coverage. However, traditional hybrid systems remain vulnerable to cold-start and sparsity problems and typically do not handle free-form natural language queries, motivating more adaptive and flexible methods that can combine structured signals with contextual understanding

2.2 Deep Learning Based Recommendation Systems

Deep learning has enabled richer feature representation and stronger predictive performance in recommender systems. Sridhar et al. (2023) proposed a content-based movie recommender that couples Deep Belief Networks (DBN) with Monarch Butterfly Optimization (MBO) for feature selection and classification; the MBO selects informative features while DBN performs deep representation learning and prediction, producing competitive MAE/RMSE and precision/recall on benchmark datasets. While such deep architectures achieve strong accuracy, they often rely on static metadata and may produce repetitive suggestions for users with narrow histories. Additionally, many deep models lack transparent, easily interpretable reasoning, which constrains explainability and user trust.

2.3 Generative AI in Recommendation Systems

Generative AI has introduced new opportunities for personalization and user engagement. Sridhar (2023) proposed a four-stage architecture, Item Retrieval, Item Mapping, Ranking, and Rendering, that integrated external metadata (IMDB) and visual assets (OMDB) to deliver thematic and mood-based recommendations. Similarly, Ayemowa et al. (2024) conducted a systematic review of 52 studies on generative AI recommenders, showing that models such as GANs, VAEs, and autoencoders outperformed classical methods by addressing cold-start, sparsity, and diversity challenges. These studies highlight the potential of generative models to enrich recommendation pipelines. However, challenges such as computational cost, latency, and limited explainability continue to restrict their scalability in real-world applications.

2.4 Large Language Models for Recommendation

2.4.1 Conversational and Interactive Recommenders

Large Language Models (LLMs) have introduced new capabilities for processing free-form user intent and producing narrative-driven recommendations. In user studies, LLM-based conversational recommenders demonstrate superior engagement and richer explanations compared with classical recommenders, yet they can underperform on personalization and diversity metrics when used alone (Sun et al., 2024). This indicates that LLMs are powerful at capturing contextual and narrative intent but often need to be integrated with structured signals to produce robust, personalized recommendations.

2.4.2 Addressing Bias in LLM-Based Systems

Lichtenberg et al. (2023) demonstrated that LLM-driven recommenders often reproduce popularity bias due to imbalances in training data. Prompt engineering reduced this bias to some extent, but deeper structural strategies remain necessary to ensure diversity and fairness in recommendations.

2.4.3 Generative Recommendation with LLMs

Ji et al. (2023) introduced GenRec, a generative recommendation framework using the LLaMA model. Unlike ranking-based methods, GenRec directly generates target items from user–item interaction sequences. While this approach achieved superior performance on datasets such as MovieLens 25M, it faced high computational demands and interpretability issues, limiting its scalability and transparency.

2.4.4 Narrative-Driven Recommendations

Eberhard et al. (2024) explored narrative-driven recommenders using LLMs to process free-form queries (e.g., “a thriller like *Shutter Island*”). Evaluating 38 open- and closed-source models, they found that LLMs outperformed traditional baselines such as doc2vec. A simple zero-shot prompting produced effective results, underscoring the robustness of LLMs in narrative-driven recommendation tasks.

The following table summarizes the key literature and the various approaches to LLM based recommendation systems.

Table 2.0 Related works

Authors & Year	Technique / Method	Advantages	Disadvantages
Mahmood et al. (2022)	Hybrid: Collaborative Filtering (CF) + Association Rule Mining	Combines user–item interactions with hidden associations, improving accuracy and coverage.	Struggles with cold-start and sparse rating matrices, limiting scalability.
Sridhar (2023)	Generative AI pipeline using IMDB/OMDB metadata & visual assets	Captures user intent beyond ratings; scalable thematic/mood-based recommendations.	Suffers from latency and high computational cost due to API calls.
Ayemowa et al. (2024)	Systematic review of 52 GenAI recommender studies	Provides comprehensive analysis of recent trends; identifies GANs/VAEs as strong performers.	Restricted to English-language studies; may miss broader perspectives.
Sridhar et al. (2023)	Deep Belief Network (DBN) + Monarch Butterfly Optimization (MBO)	Achieves strong predictive accuracy; robust optimization improves recommendation precision/recall.	Limited personalization, privacy risks, and low diversity; scalability issues.
Siet et al. (2024)	Transformer-based sequence model + KMeans clustering	Tackles cold-start effectively; enables high personalization and sequence modeling.	Computationally intensive and prone to overfitting in sparse datasets.
Ji et al. (2023)	GenRec: Generative Recommender with LLaMA model	Directly generates items from user histories; provides novel	Requires high GPU resources; dataset-

		personalization beyond ranking.	dependent; lacks interpretability.
Sun et al. (2024)	LLMs as conversational recommenders (LLMRec vs MovieRec)	Improves engagement and explainability through natural conversation.	Weak in personalization, novelty, and diversity compared to traditional baselines.
Eberhard et al. (2024)	Narrative-driven LLM recommendations with zero-shot prompting	Handles free-form queries effectively; simple prompts yield strong results.	Limited to specific narrative tasks; lacks robustness in broader domains.
Mu & Wu (2023)	Multimodal recommender (text + visual + deep learning)	Enhances personalization by integrating visual/textual features; reduces sparsity.	Visual data contributes marginally; sparsity remains unsolved.
Lichtenberg et al. (2023)	Study on LLM-driven popularity bias	Introduces new bias metrics; shows LLMs reduce bias compared to traditional models.	Evaluation simplified; mitigation strategies underdeveloped; bias persists.

2.5. Limitations in Existing Researches

Despite notable progress, existing research on generative AI and LLM-powered recommender systems presents several limitations that warrant further investigation. First, long-standing challenges such as the cold-start problem and data sparsity remain only partially addressed. While generative models such as GANs and VAEs demonstrate potential in mitigating these issues, they continue to struggle when recommending items for entirely new users or for niche content with minimal historical interaction data (Ayemowa et al., 2024; Ji et al., 2023).

Secondly, personalization in LLM-based recommenders remains incomplete. Although these models excel at contextual understanding and conversational engagement, studies consistently show that they underperform in delivering highly individualized recommendations, particularly for users with complex or extensive viewing histories (Sun et al., 2024; Eberhard et al., 2024).

Another recurring concern is bias, especially popularity bias, where models disproportionately promote mainstream content due to imbalances in training data. While prompt engineering has been applied as a partial solution, structural strategies for ensuring diversity and fairness are still underdeveloped (Lichtenberg et al., 2023).

Practical deployment also poses significant challenges. The computational and resource demands of training and inference for large-scale models create scalability and latency constraints, limiting their feasibility for real-time applications in operational settings (Ji et al., 2023).

Furthermore, explainability remains a critical weakness. Although LLMs can generate natural language justifications, these explanations are often superficial, ungrounded, or inconsistent, raising concerns about transparency and user trust (Eberhard et al., 2024). Large-scale, longitudinal, and cross-domain evaluations remain rare, leaving a gap between academic advances and practical deployment (Ayemowa et al., 2024; Sridhar, 2023).

2.6. Future Work and Research Opportunities

Given these limitations, several avenues for future research and development emerge. One promising direction lies in the integration of LLMs with traditional or hybrid recommendation approaches, combining the contextual reasoning and fluency of LLMs with the scalability and robustness of classical models. Such hybrid frameworks may provide a more balanced trade-off between personalization, novelty, diversity, and computational efficiency. Addressing bias represents another critical research frontier. Beyond mitigating popularity bias, future systems should embed fairness-aware mechanisms to ensure equitable recommendations across demographic, cultural, and linguistic contexts, thereby enhancing inclusivity and reducing systemic exposure imbalances.

Equally important are efforts to improve the practical scalability of LLM-powered recommenders. Research into parameter-efficient tuning, model compression, and lightweight inference techniques could reduce computational overhead, making these models viable for real-time applications and even edge deployments. Additionally, the integration of multimodal and narrative-driven inputs, such as textual descriptions, visual content, and audio cues which offers a promising pathway for capturing the full richness of user intent and delivering highly adaptive, context-aware recommendations.

Finally, explainability and evaluation remain areas requiring significant advancement. Future systems should generate grounded, evidence-based explanations that articulate the reasoning behind recommendations, thereby fostering user trust and transparency. Moreover, longitudinal and cross-domain evaluations should be prioritized, moving beyond benchmark datasets to real-world operational environments such as streaming platforms, e-commerce, and education. Together, these directions highlight the potential for a new generation of recommender systems that are not only accurate and conversational but also fair, explainable, scalable, and user-centric, bridging the gap between academic innovation and practical utility.

CHAPTER 3: METHODOLOGY

This study employs the OSEMN framework that is **Obtain, Scrub, Explore, Model, and Interpret**, to design and implement a hybrid recommendation framework. The OSEMN approach provides a structured and widely recognized methodology for data science research, ensuring systematic data handling, rigorous modeling, and coherent evaluation. The research methodology is closely aligned with the stated objectives: (i) to analyze key features influencing predictive modeling, (ii) to develop a hybrid recommendation framework integrating predictive modeling with LLM-based query interpretation, and (iii) to evaluate the framework through comparative analysis using standard recommendation metrics.

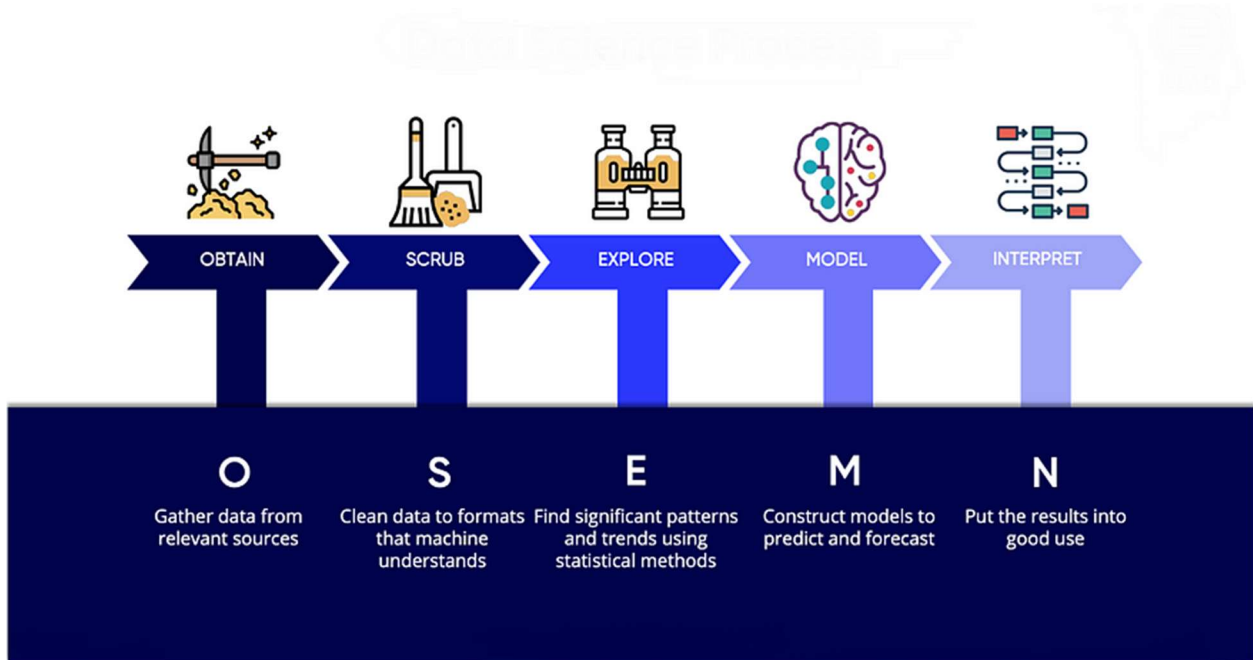


Figure 3.1: Research Design

3.1 Obtain

The first stage involves acquiring the dataset required for this research. The study utilizes the *Movies dataset* compiled by Abdallah Wagih Ibrahim and made publicly available on Kaggle. This dataset aggregates metadata from multiple sources and contains approximately 4,800 entries covering a broad spectrum of genres, time periods, and regions. It incorporates both structured variables, such as budget, revenue, runtime, genres, cast, crew, director, vote counts, ratings, and popularity and unstructured fields including overview, tagline, and keywords. The availability of both structured and textual features makes the dataset suitable for hybrid recommendation tasks that combine predictive modeling with natural language interpretation.

For the purposes of this research, two target variables are identified: **vote_average**, representing aggregated user ratings, and **popularity**, reflecting audience engagement. These targets allow the framework to model both subjective assessments of quality and broader measures of audience

interest, thus aligning with the dual objectives of personalization and engagement in recommender systems.

3.2 Scrub

The second stage focuses on data cleaning and preprocessing to ensure the dataset is accurate, consistent, and analytically usable. Numerical variables such as budget, revenue, and runtime, which often contain missing or inconsistent values, are treated through appropriate imputation strategies, while categorical features such as cast and director are marked as *Unknown* when unavailable. To enable computational analysis, categorical variables including genres, cast, and director are transformed using multi-hot encoding or embedding-based representations. Temporal refinement is achieved by extracting release year from the `release_date` attribute, thereby allowing more meaningful analysis of temporal patterns. Textual attributes, such as overview, tagline, and keywords, undergo preprocessing steps including tokenization, stop-word removal, and lemmatization, followed by numerical representation using TF-IDF or transformer-based embeddings. Additional feature engineering generates derived attributes, such as sentiment polarity from plot descriptions, rating bins to classify movies into high and low categories, and genre-level popularity measures. These processes collectively ensure that the dataset is not only clean and consistent but also sufficiently enriched to support robust predictive modeling and recommendation.

3.3 Explore

The third stage involves exploratory data analysis (EDA), which provides descriptive and inferential insights into the dataset and prepares for the modeling phase. Statistical summaries are computed to examine measures of central tendency and variability across numerical attributes, including budget, revenue, runtime, ratings, and popularity. Distributional analysis highlights the skewness and outliers that characterize these features, offering a deeper understanding of their variability.

Correlation analysis using Pearson and Spearman coefficients is applied to uncover dependencies between variables, such as the association between vote counts and popularity or between budget and revenue. Genre-based and temporal analyses are conducted to investigate how ratings and popularity differ across genres and change over time. Visualization techniques include scatterplots, bar charts, and heatmaps, to support interpretability by presenting these relationships in a clear and accessible form. This stage plays a crucial role in identifying the features most influential in predicting popularity and user ratings, thereby directly addressing the first research objective.

3.4 Model

The modeling stage forms the core methodological contribution of this study and integrates predictive modeling, LLM-based query interpretation, generative AI enrichment, and hybrid recommendation into a unified framework.

The first component is predictive modeling, where algorithms such as Random Forest and XGBoost are trained to forecast movie ratings and popularity. These models are evaluated using metrics including RMSE, accuracy, precision, recall, and F1-score. The predictive outputs serve as structured signals that provide a foundation for the hybrid recommendation system.

The second component is LLM-based query interpretation, which enables the system to process free-form natural language queries. Large Language Models, such as GPT-4, Gemini and LLaMA-2, are employed to translate narrative-driven requests into structured preference profiles through techniques such as entity recognition and sentiment analysis.

The third component is generative AI enrichment, which addresses gaps in metadata and enhances recommendation transparency. Generative models are employed to impute missing values (e.g., absent genres) and to produce enhanced textual summaries. Furthermore, generative AI is used to create natural language explanations that clarify the rationale behind recommendations, thereby increasing system interpretability and user trust.

The final component is the hybrid recommendation framework, which integrates predictive modeling outputs, content-based similarity measures, and popularity-based filtering through a weighted fusion strategy. To mitigate popularity bias and enhance novelty, diversity-promoting techniques such as niche boosting are incorporated. The result is a framework that balances accuracy, diversity, and explainability, thus addressing the second research objective.

3.5 Interpret

The final stage focuses on the evaluation and deployment of the proposed framework. System performance is assessed using established recommendation metrics. Accuracy is measured using Precision@K and Recall@K, while broader system effectiveness is evaluated through novelty, diversity, and long-tail bias indicators. Comparative analysis is carried out against baseline models, including collaborative filtering, content-based filtering, and LLM-only approaches, to assess the relative performance gains of the hybrid framework.

To validate its practical feasibility, a prototype system is developed. The backend is implemented using FastAPI or Flask, integrating predictive models with LLM-based query interpretation, while the frontend is built using Streamlit to provide an interactive and user-friendly recommendation interface. This feature enhances system transparency and fosters user trust.

The Interpret stage thus ensures that the framework is evaluated with academic rigor while also demonstrating its applicability in real-world contexts. By uniting systematic evaluation with prototype deployment, the research fulfils its third objective and contributes toward building a more accurate, explainable, and user-centric recommendation system.

3.6 Conclusion

This research proposes a hybrid movie recommendation framework developed through the OSEMN methodology, integrating predictive modeling, traditional filtering techniques, and LLM-powered query interpretation. The framework extends beyond accuracy to prioritize novelty, diversity, and explainability, thereby enhancing user trust and long-term engagement. By combining predictive baselines with context-aware natural language understanding and generative enrichment, it seeks to address persistent challenges in recommender systems, including cold-start limitations, popularity bias, and lack of transparency. The deployment of a functional prototype further demonstrates the practical feasibility of the proposed system, while its outcomes are anticipated to contribute to the advancement of next-generation recommendation models that are not only accurate and personalized but also fair, explainable, and adaptable across domains such as e-commerce, education, and digital media.

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Dataset link: <https://www.kaggle.com/datasets/abdallahwagih/movies/data>